

พื้นฐานระบบฐานข้อมูล ER-Model

ฐานข้อมูล(Database) : Northwind ตัวอย่างข้อมูลดังนี้

ตาราง : Customers

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taqueria	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

ตาราง : Employees

EmployeeID	LastName	FirstName	BirthDate	Photo	Notes
1	Davolio	Nancy	12/8/1968	EmpID1.pic	Education includes a BA in psychology from Colorado State University. She also completed (The Art of the Cold Call). Nancy is a member of Toastmasters International.
2	Fuller	Andrew	2/19/1952	EmpID2.pic	Andrew received his BTS commercial and a Ph.D. in international marketing from the University of Dallas. He is fluent in French and Italian and reads German. He joined the company as a sales representative, was promoted to sales
3	Leverling	Janet	8/30/1963	EmpID3.pic	Janet has a BS degree in chemistry from Boston College. She has also completed a certificate program in food retailing management. Janet was hired as a sales associate and was promoted to sales representative.

ตาราง : Shippers

ShipperID	ShipperName	Phone
1	Speedy Express	(503) 555-9831
2	United Package	(503) 555-3199
3	Federal Shipping	(503) 555-9931

ตาราง : Suppliers

SupplierID	SupplierName	ContactName	Address	City	PostalCode	Country	Phone
1	Exotic Liquid	Charlotte Cooper	49 Gilbert St.	Londona	EC1 4SD	UK	(171) 555-2222
2	New Orleans Cajun Delights	Shelley Burke	P.O. Box 78934	New Orleans	70117	USA	(100) 555-4822
3	Grandma Kelly's Homestead	Regina Murphy	707 Oxford Rd.	Ann Arbor	48104	USA	(313) 555-5735

ตาราง : Categories

CategoryID	CategoryName	Description
1	Beverages	Soft drinks, coffees, teas, beers, and ales
2	Condiments	Sweet and savory sauces, relishes, spreads, and seasonings
3	Confections	Desserts, candies, and sweet breads

ตาราง : Products

ProductID	ProductName	ShipperID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25

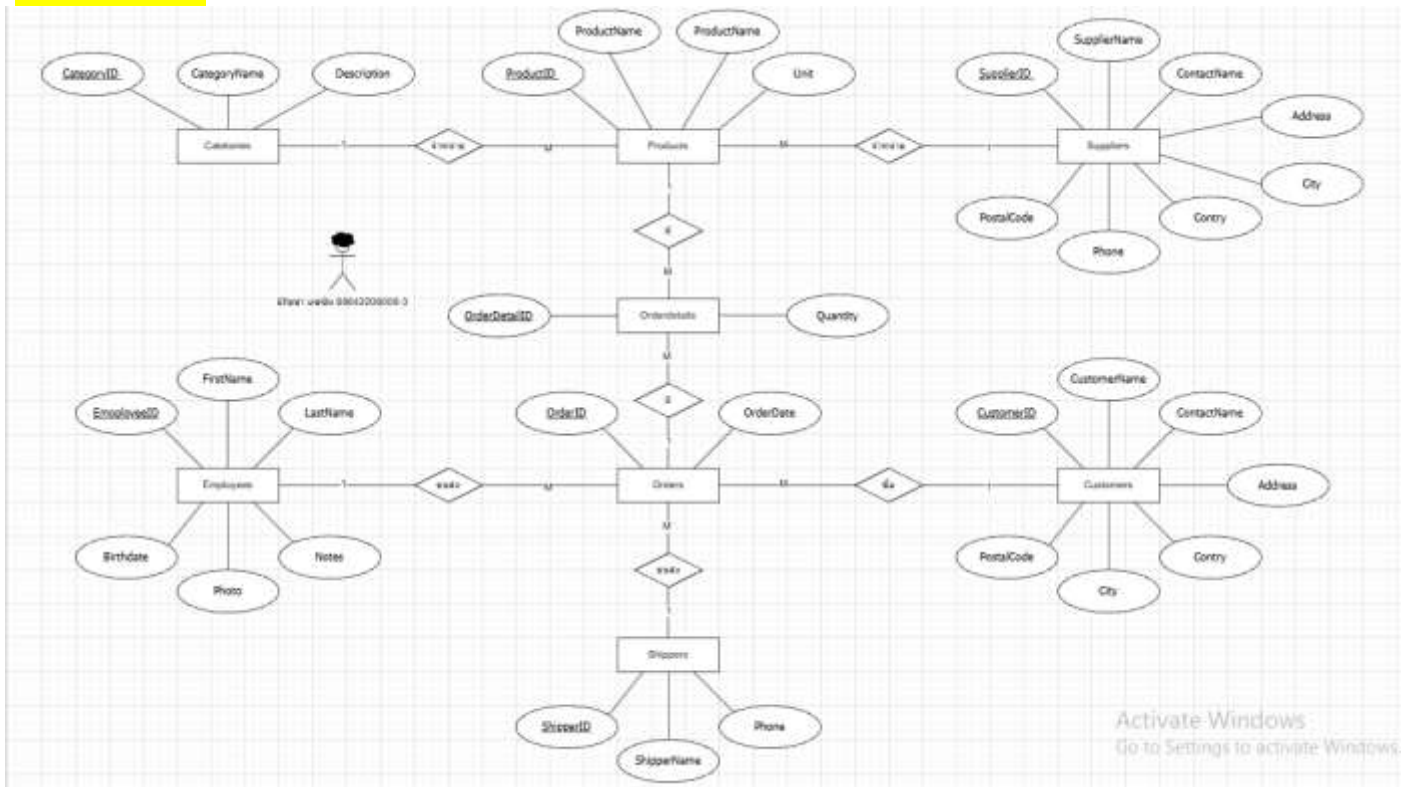
ตาราง : Orders

OrderID	CustomerID	EmployeeID	OrderDate	ShipperID
10248	2	1	7/4/1996	3
10249	1	1	7/5/1996	1
10250	1	2	7/8/1996	2
10251	2	3	7/8/1996	1
10252	3	2	7/9/1996	2
10253	2	1	7/10/1996	2

ตาราง : OrderDetail

OrderDetailID	OrderID	ProductID	Quantity
1	10248	1	12
2	10248	2	10
3	10248	3	5
4	10249	2	9
5	10249	3	40
6	10250	1	10
7	10250	2	35
8	10250	3	15

1.ให้เขียนแผนภาพ ER-Model แบบ Chain's [การออกแบบระดับความคิด Conceptual Database Design]
 [Visio : draw.IO]



2.จากแผนภาพ ER-Model [ออกแบบระดับตรรกะ Logical Database Design]

Categories(CategoryID , CategoryName, Description)

Suppliers(SupplierID , SupplierName, ContactName, Address , City,
 PostalCode, Country, Phone)

Products(ProductID , ProductName, SupplierID , CategoryID , Unit , Price)

Customers(CustomerID, CustomerName, ContactName, Address, City, PostalCode,Country)

Employees(EmployeeID, FirstName, LastName, BirthDate, Photo, Notes)

Orders(OrderID, OrderDate, CustomerID, EmployeeID, ShipperID)

Shippers(ShipperID, ShipperName, Phone)

OrderDetails(OrderDetailID, OrderID, Quantity , ProductID)

3. ออกแบบระดับกายภาพ Physical database design

Table Name	Attribute Name	Description	Datatype	Size	Key	References
Categories	CategoryID	รหัสหมวดสินค้า	int	11	PK	
	CategoryName	ชื่อหมวด	Varchar	50		
	Description	อธิบายเพิ่มเติม	Varchar	100		
Suppliers	SupplierID	รหัสผู้ผลิตสินค้า	Int	11	PK	
	SupplierName	ชื่อบริษัทผลิตสินค้า	Varchar	100		
	ContactName	ชื่อผู้ติดต่อ	Varchar	100		
	Address	ที่อยู่	Varchar	255		
	City	เมือง	Varchar	100		
	PostalCode	รหัสไปรษณีย์	Varchar	20		
	Country	จังหวัด	Varchar	100		
	Phone	เบอร์โทร	Varchar	20		
Products	ProductID	รหัสสินค้า	Int	11	PK	
	ProductName	ชื่อสินค้า	Varchar	100		
	SupplierID	รหัสผู้ผลิตสินค้า	Int	11	Fk	Suppliers
	CategoryID	รหัสหมวดสินค้า	Int	11	Fk	Categories
	Unit	หน่วยสินค้า	Varchar	50		
	Price	ราคาต่อหน่วย	Decimal	8.2		
Customers	CustomerID	รหัสลูกค้า	int	11	PK	
	CustomerName	ชื่อลูกค้า	Varchar	100		
	ContactName	ชื่อผู้ติดต่อ	Varchar	100		
	Address	ที่อยู่	Varchar	255		
	City	เมือง	Varchar	100		
	PostalCode	รหัสไปรษณีย์	Varchar	20		
	Country	จังหวัด	Varchar	100		
Orders	OrderID	รหัสการสั่งซื้อ	int	11	PK	
	CustomerID	รหัสลูกค้า	int	11	FK	Customer
	EmployeeID	รหัสพนักงาน	int	11	FK	Employee
	OrderDate	วันที่สั่งซื้อ	Date	11		
	ShipperID	รหัสบริษัทขนส่ง	int	11	FK	Shipper
OrderDetails	OrderDetailID	รหัสรายละเอียดการสั่งซื้อ	int	11	PK	
	OrderID	รหัสการสั่งซื้อ	int	11	FK	Orders
	ProductID	รหัสสินค้า	int	11	FK	Products
	Quantity	จำนวนการที่สั่งซื้อ	int	11		

Employees	EmployeeID	รหัสพนักงาน	Int	11	PK	
	FirstName	ชื่อพนักงาน	Varchar	100		
	LastName	นามสกุลพนักงาน	Varchar	100		
	BirthDate	วันเกิด	Date			
	Photo	รูปพนักงาน	Varchar	255		
	Notes	หมายเหตุเพิ่มเติม	text			
Shippers	ShipperID	รหัสบริษัทขนส่ง	Int	11	PK	
	ShipperName	ชื่อบริษัทขนส่ง	Varchar	100		
	Phone	เบอร์โทรศัพท์	Varchar	20		

เพิ่มอีก 5 Table

4. แสดงข้อมูลทั้งหมด ตาราง Customers เรียงลำดับ CustomerID มากไปหาน้อย

SELECT *

FROM customers

ORDER BY CustomerID DESC;

5. แสดงข้อมูล CustomerID CustomerName PostalCode ตาราง Customers เงื่อนไขเฉพาะ Country เป็น Mexico

SELECT CustomerID, CustomerName, PostalCode

FROM customers

WHERE Country = 'Mexico';

6. แสดงข้อมูลทั้งหมด ตาราง Products เงื่อนไข price มากกว่าหรือเท่ากับ 19 และ ShipperID เท่ากับ 1

SELECT *

FROM products

WHERE Price >= 19

AND SupplierID = 1;

7. จากข้อ 6 ผลลัพธ์ คือ

←T→	ProductID	ProductName	SupplierID	CategoryID	Unit	Price
☐ Edit ☐ Copy ☐ Delete	2 Chang		1	1	24 - 12 oz bottles	19.00

8. ผลลัพธ์คำสั่ง Select Distince ShipperID from Products Order by ShipperID Desc

✓ Showing rows 0 - 24 (29 total, Query took 0.0004 seconds.) [SupplierID: 29... - 5...]

```
Select DISTINCT SupplierID from products Order by SupplierID Desc;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

1 > >> | ☐ Show all | Number of rows: 25 Filter rows: Search this table Sort by key:

Extra options

SupplierID
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9. แสดงข้อมูลเชื่อมความสัมพันธ์ 2 ตาราง คือ Products และ Categories โดยแสดงข้อมูล ProductID ProductName CategoryID CategoryName Price โดยเรียงลำดับ CategoryID น้อยไปหามาก

```
SELECT  
p.ProductID,  
p.ProductName,
```

```

c.CategoryID,
c.CategoryName,
p.Price
FROM products p
INNER JOIN categories c
ON p.CategoryID = c.CategoryID
ORDER BY c.CategoryID ASC;

```

10.ถ้าต้องการผลลัพธ์ ต้องการทราบจำนวนรายการสินค้า ของแต่ละ Category ดังนี้

CategoryID	CategoryName	CountNum
2	Condiments	4
1	Beverages	2

คำสั่งภาษา SQL :

```

SELECT
c.CategoryID,
c.CategoryName,
COUNT(p.ProductID) AS CountNum
FROM categories c
INNER JOIN products p
ON c.CategoryID = p.CategoryID
GROUP BY c.CategoryID, c.CategoryName
ORDER BY c.CategoryID DESC;

```

11. สร้างวิว(View) มุมมองข้อมูล ชื่อ View_Products_All เชื่อม 3 ตาราง คือ Products Categories และ Suppliers โดยแสดงข้อมูลคือข้อมูล ProductID ProductName CategoryID CategoryName Price SupplierID SupplierName ContractName Phone

```

CREATE VIEW View_Products_All AS
SELECT
p.ProductID,
p.ProductName,
c.CategoryID,
c.CategoryName,

```

```

p.Price,
s.SupplierID,
s.SupplierName,
s.ContactName,
s.Phone
FROM products p
INNER JOIN categories c
  ON p.CategoryID = c.CategoryID
INNER JOIN suppliers s
  ON p.SupplierID = s.SupplierID;

```

12. จากตาราง Order ต้องการทราบ OrderID OrderDate CustomerName LastName FirstName และ ShipperName Phone ใช้คำสั่ง SQL เชื่อมความสัมพันธ์กัน 4 ตาราง คือ Orders Customers Employee และ Shippers

```

SELECT
  o.OrderID,
  o.OrderDate,
  c.CustomerName,
  e.LastName,
  e.FirstName,
  s.ShipperName,
  s.Phone
FROM orders o
INNER JOIN customers c
  ON o.CustomerID = c.CustomerID
INNER JOIN employees e
  ON o.EmployeeID = e.EmployeeID
INNER JOIN shippers s
  ON o.ShipperID = s.ShipperID;

```

13. จากตาราง OrderDetail ใบเสร็จ OrderID ที่ 10248 แสดงข้อมูล OrderID ProductName Quantity ใช้คำสั่งภาษา SQL :

```
SELECT
    o.OrderID,
    p.ProductName,
    od.Quantity
FROM orderdetails od
INNER JOIN orders o
    ON od.OrderID = o.OrderID
INNER JOIN products p
    ON od.ProductID = p.ProductID
WHERE o.OrderID = 10248;
```

ตัวอย่างฐานข้อมูล

ฐานข้อมูล : **db_northwind**

```
-- Database: `db_northwind`

-- Table structure for table `categories`
CREATE TABLE `categories` (
  `CategoryID` int(11) NOT NULL COMMENT 'รหัสประเภทสินค้า',
  `CategoryName` varchar(50) DEFAULT NULL COMMENT 'ชื่อประเภท',
  `Description` varchar(100) DEFAULT NULL COMMENT 'อธิบายรายละเอียด'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

--

-- Dumping data for table `categories`
--

INSERT INTO `categories` (`CategoryID`, `CategoryName`, `Description`) VALUES
(1, 'Beverages(เครื่องดื่ม)', 'Soft drinks, coffees, teas, beers, and ales'),
(2, 'Condiments', 'Sweet and savory sauces, relishes, spreads, and seasonings'),
(3, 'Confections', 'Desserts, candies, and sweet breads'),
(4, 'Dairy Products', 'Cheeses'),
(5, 'Grains/Cereals', 'Breads, crackers, pasta, and cereal'),
(6, 'Meat/Poultry', 'Prepared meats'),
```



```
(7, 'Produce', 'Dried fruit and bean curd'),  
(8, 'Seafood', 'Seaweed and fish');
```

```
-- -----
```

```
--
```

```
-- Table structure for table `customers`
```

```
--
```

```
CREATE TABLE `customers` (  
  `CustomerID` int(11) NOT NULL,  
  `CustomerName` varchar(255) DEFAULT NULL,  
  `ContactName` varchar(255) DEFAULT NULL,  
  `Address` varchar(255) DEFAULT NULL,  
  `City` varchar(100) DEFAULT NULL,  
  `PostalCode` varchar(20) DEFAULT NULL,  
  `Country` varchar(100) DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```
--
```

```
-- Dumping data for table `customers`
```

```
--
```

```
INSERT INTO `customers` (`CustomerID`, `CustomerName`, `ContactName`, `Address`, `City`,  
  `PostalCode`, `Country`) VALUES  
(1, 'Alfreds Futterkiste', 'Maria Anders', 'Obere Str. 57', 'Berlin', '12209', 'Germany'),  
(2, 'Ana Trujillo Emparedados y helados', 'Ana Trujillo', 'Avda. de la Constitución 2222', 'México D.F.', '05021', 'Mexico'),  
(3, 'Antonio Moreno Taquería', 'Antonio Moreno', 'Mataderos 2312', 'México D.F.', '05023', 'Mexico'),  
(4, 'Around the Horn', 'Thomas Hardy', '120 Hanover Sq.', 'London', 'WA1 1DP', 'UK'),  
(5, 'Berglunds snabbköp', 'Christina Berglund', 'Berguvsvägen 8', 'Luleå', 'S-958 22', 'Sweden'),  
(6, 'Blauer See Delikatessen', 'Hanna Moos', 'Forsterstr. 57', 'Mannheim', '68306', 'Germany'),  
(7, 'Blondel père et fils', 'Frédérique Citeaux', '24, place Kléber', 'Strasbourg', '67000', 'France'),  
(8, 'Bólido Comidas preparadas', 'Martín Sommer', 'C/ Araquil, 67', 'Madrid', '28023', 'Spain'),  
(9, 'Bon app\\', 'Laurence Lebihans', '12, rue des Bouchers', 'Marseille', '13008', 'France'),
```

(10, 'Bottom-Dollar Marketse', 'Elizabeth Lincoln', '23 Tsawassen Blvd.', 'Tsawassen', 'T2F 8M4', 'Canada'),

(11, 'B's Beverages', 'Victoria Ashworth', 'Fauntleroy Circus', 'London', 'EC2 5NT', 'UK'),

(12, 'Cactus Comidas para llevar', 'Patricio Simpson', 'Cerrito 333', 'Buenos Aires', '1010', 'Argentina'),

(13, 'Centro comercial Moctezuma', 'Francisco Chang', 'Sierras de Granada 9993', 'México D.F.', '05022', 'Mexico'),

(14, 'Chop-suey Chinese', 'Yang Wang', 'Hauptstr. 29', 'Bern', '3012', 'Switzerland'),

(15, 'Comércio Mineiro', 'Pedro Afonso', 'Av. dos Lusíadas, 23', 'São Paulo', '05432-043', 'Brazil'),

(16, 'Consolidated Holdings', 'Elizabeth Brown', 'Berkeley Gardens 12 Brewery', 'London', 'WX1 6LT', 'UK'),

(17, 'Drachenblut Delikatessend', 'Sven Ottlieb', 'Walserweg 21', 'Aachen', '52066', 'Germany'),

(18, 'Du monde entier', 'Janine Labrune', '67, rue des Cinquante Otages', 'Nantes', '44000', 'France'),

(19, 'Eastern Connection', 'Ann Devon', '35 King George', 'London', 'WX3 6FW', 'UK'),

(20, 'Ernst Handel', 'Roland Mendel', 'Kirchgasse 6', 'Graz', '8010', 'Austria'),

(21, 'Familia Arquibaldo', 'Aria Cruz', 'Rua Orós, 92', 'São Paulo', '05442-030', 'Brazil'),

(22, 'FISSA Fabrica Inter. Salchichas S.A.', 'Diego Roel', 'C/ Morazarzal, 86', 'Madrid', '28034', 'Spain'),

(23, 'Folies gourmandes', 'Martine Rancé', '184, chaussée de Tournai', 'Lille', '59000', 'France'),

(24, 'Folk och få HB', 'Maria Larsson', 'Åkergatan 24', 'Bräcke', 'S-844 67', 'Sweden'),

(25, 'Frankenversand', 'Peter Franken', 'Berliner Platz 43', 'München', '80805', 'Germany'),

(26, 'France restauration', 'Carine Schmitt', '54, rue Royale', 'Nantes', '44000', 'France'),

(27, 'Franchi S.p.A.', 'Paolo Accorti', 'Via Monte Bianco 34', 'Torino', '10100', 'Italy'),

(28, 'Furia Bacalhau e Frutos do Mar', 'Lino Rodriguez', 'Jardim das rosas n. 32', 'Lisboa', '1675', 'Portugal'),

(29, 'Galería del gastrónomo', 'Eduardo Saavedra', 'Rambla de Cataluña, 23', 'Barcelona', '08022', 'Spain'),

(30, 'Godos Cocina Típica', 'José Pedro Freyre', 'C/ Romero, 33', 'Sevilla', '41101', 'Spain'),

(31, 'Gourmet Lanchonetes', 'André Fonseca', 'Av. Brasil, 442', 'Campinas', '04876-786', 'Brazil'),

(32, 'Great Lakes Food Market', 'Howard Snyder', '2732 Baker Blvd.', 'Eugene', '97403', 'USA'),

(33, 'GROSELLA-Restaurante', 'Manuel Pereira', '5ª Ave. Los Palos Grandes', 'Caracas', '1081', 'Venezuela'),

(34, 'Hanari Carnes', 'Mario Pontes', 'Rua do Paço, 67', 'Rio de Janeiro', '05454-876', 'Brazil'),

(35, 'HILARIÓN-Abastos', 'Carlos Hernández', 'Carrera 22 con Ave. Carlos Soublette #8-35', 'San Cristóbal', '5022', 'Venezuela'),

(36, 'Hungry Coyote Import Store', 'Yoshi Latimer', 'City Center Plaza 516 Main St.', 'Elgin', '97827', 'USA'),

(37, 'Hungry Owl All-Night Grocers', 'Patricia McKenna', '8 Johnstown Road', 'Cork', '', 'Ireland'),

(38, 'Island Trading', 'Helen Bennett', 'Garden House Crowther Way', 'Cowes', 'PO31 7PJ', 'UK'),

(39, 'Königlich Essen', 'Philip Cramer', 'Maubelstr. 90', 'Brandenburg', '14776', 'Germany'),

(40, 'La corne d'abondance', 'Daniel Tonini', '67, avenue de l'Europe', 'Versailles', '78000', 'France'),

(41, 'La maison d'Asie', 'Annette Roulet', '1 rue Alsace-Lorraine', 'Toulouse', '31000', 'France'),

(42, 'Laughing Bacchus Wine Cellars', 'Yoshi Tannamuri', '1900 Oak St.', 'Vancouver', 'V3F 2K1', 'Canada'),

(43, 'Lazy K Kountry Store', 'John Steel', '12 Orchestra Terrace', 'Walla Walla', '99362', 'USA'),

(44, 'Lehmanns Marktstand', 'Renate Messner', 'Magazinweg 7', 'Frankfurt a.M.', '60528', 'Germany'),

(45, 'Let's Stop N Shop', 'Jaime Yorres', '87 Polk St. Suite 5', 'San Francisco', '94117', 'USA'),

(46, 'LILA-Supermercado', 'Carlos González', 'Carrera 52 con Ave. Bolívar #65-98 Llano Largo', 'Barquisimeto', '3508', 'Venezuela'),

(47, 'LINO-Delicateses', 'Felipe Izquierdo', 'Ave. 5 de Mayo Porlamar', 'I. de Margarita', '4980', 'Venezuela'),

(48, 'Lonesome Pine Restaurant', 'Fran Wilson', '89 Chiaroscuro Rd.', 'Portland', '97219', 'USA'),

(49, 'Magazzini Alimentari Riuniti', 'Giovanni Rovelli', 'Via Ludovico il Moro 22', 'Bergamo', '24100', 'Italy'),

(50, 'Maison Dewey', 'Catherine Dewey', 'Rue Joseph-Bens 532', 'Bruxelles', 'B-1180', 'Belgium'),

(51, 'Mère Paillarde', 'Jean Fresnière', '43 rue St. Laurent', 'Montréal', 'H1J 1C3', 'Canada'),

(52, 'Morgenstern Gesundkost', 'Alexander Feuer', 'Heerstr. 22', 'Leipzig', '04179', 'Germany'),

(53, 'North/South', 'Simon Crowther', 'South House 300 Queensbridge', 'London', 'SW7 1RZ', 'UK'),

(54, 'Océano Atlántico Ltda.', 'Yvonne Moncada', 'Ing. Gustavo Moncada 8585 Piso 20-A', 'Buenos Aires', '1010', 'Argentina'),

(55, 'Old World Delicatessen', 'Rene Phillips', '2743 Bering St.', 'Anchorage', '99508', 'USA'),

(56, 'Ottilies Käseladen', 'Henriette Pfalzheim', 'Mehrheimerstr. 369', 'Köln', '50739', 'Germany'),

(57, 'Paris spécialités', 'Marie Bertrand', '265, boulevard Charonne', 'Paris', '75012', 'France'),

(58, 'Pericles Comidas clásicas', 'Guillermo Fernández', 'Calle Dr. Jorge Cash 321', 'México D.F.', '05033', 'Mexico'),

(59, 'Piccolo und mehr', 'Georg Pipps', 'Geislweg 14', 'Salzburg', '5020', 'Austria'),

(60, 'Princesa Isabel Vinhoss', 'Isabel de Castro', 'Estrada da saúde n. 58', 'Lisboa', '1756', 'Portugal'),

(61, 'Que Delícia', 'Bernardo Batista', 'Rua da Panificadora, 12', 'Rio de Janeiro', '02389-673', 'Brazil'),

(62, 'Queen Cozinha', 'Lúcia Carvalho', 'Alameda dos Canários, 891', 'São Paulo', '05487-020', 'Brazil'),

(63, 'QUICK-Stop', 'Horst Kloss', 'Taucherstraße 10', 'Cunewalde', '01307', 'Germany'),

(64, 'Rancho grande', 'Sergio Gutiérrez', 'Av. del Libertador 900', 'Buenos Aires', '1010', 'Argentina'),

(65, 'Rattlesnake Canyon Grocery', 'Paula Wilson', '2817 Milton Dr.', 'Albuquerque', '87110', 'USA'),

(66, 'Reggiani Caseifici', 'Maurizio Moroni', 'Strada Provinciale 124', 'Reggio Emilia', '42100', 'Italy'),

(67, 'Ricardo Adocicados', 'Janete Limeira', 'Av. Copacabana, 267', 'Rio de Janeiro', '02389-890', 'Brazil'),

(68, 'Richter Supermarkt', 'Michael Holz', 'Grenzacherweg 237', 'Genève', '1203', 'Switzerland'),

(69, 'Romero y tomillo', 'Alejandra Camino', 'Gran Vía, 1', 'Madrid', '28001', 'Spain'),

(70, 'Santé Gourmet', 'Jonas Bergulfsen', 'Erling Skakkes gate 78', 'Stavern', '4110', 'Norway'),

(71, 'Save-a-lot Markets', 'Jose Pavarotti', '187 Suffolk Ln.', 'Boise', '83720', 'USA'),

(72, 'Seven Seas Imports', 'Hari Kumar', '90 Wadhurst Rd.', 'London', 'OX15 4NB', 'UK'),

(73, 'Simons bistro', 'Jytte Petersen', 'Vinbæltet 34', 'København', '1734', 'Denmark'),

(74, 'Spécialités du monde', 'Dominique Perrier', '25, rue Lauriston', 'Paris', '75016', 'France'),

(75, 'Split Rail Beer & Ale', 'Art Braunschweiger', 'P.O. Box 555', 'Lander', '82520', 'USA'),

(76, 'Suprêmes délices', 'Pascale Cartrain', 'Boulevard Tirou, 255', 'Charleroi', 'B-6000', 'Belgium'),

(77, 'The Big Cheese', 'Liz Nixon', '89 Jefferson Way Suite 2', 'Portland', '97201', 'USA'),

(78, 'The Cracker Box', 'Liu Wong', '55 Grizzly Peak Rd.', 'Butte', '59801', 'USA'),

(79, 'Toms Spezialitäten', 'Karin Josephs', 'Luisenstr. 48', 'Münster', '44087', 'Germany'),

(80, 'Tortuga Restaurante', 'Miguel Angel Paolino', 'Avda. Azteca 123', 'México D.F.', '05033', 'Mexico'),

(81, 'Tradição Hipermercados', 'Anabela Domingues', 'Av. Inês de Castro, 414', 'São Paulo', '05634-030', 'Brazil'),

(82, 'Trail\'s Head Gourmet Provisioners', 'Helvetius Nagy', '722 DaVinci Blvd.', 'Kirkland', '98034', 'USA'),

(83, 'Vaffeljernet', 'Palle Ibsen', 'Smagsløget 45', 'Århus', '8200', 'Denmark'),

(84, 'Victuailles en stock', 'Mary Saveley', '2, rue du Commerce', 'Lyon', '69004', 'France'),

(85, 'Vins et alcools Chevalier', 'Paul Henriot', '59 rue de l\'Abbaye', 'Reims', '51100', 'France'),

(86, 'Die Wandernde Kuh', 'Rita Müller', 'Adenauerallee 900', 'Stuttgart', '70563', 'Germany'),

(87, 'Wartian Herkku', 'Pirkko Koskitalo', 'Torikatu 38', 'Oulu', '90110', 'Finland'),
(88, 'Wellington Importadora', 'Paula Parente', 'Rua do Mercado, 12', 'Resende', '08737-363',
'Brazil'),
(89, 'White Clover Markets', 'Karl Jablonski', '305 - 14th Ave. S. Suite 3B', 'Seattle', '98128',
'USA'),
(90, 'Wilman Kala', 'Matti Karttunen', 'Keskuskatu 45', 'Helsinki', '21240', 'Finland'),
(91, 'Wolski', 'Zbyszek', 'ul. Filtrowa 68', 'Walla', '01-012', 'Poland');

-- -----

--

-- Table structure for table `employees`

--

```
CREATE TABLE `employees` (  
  `EmployeeID` int(11) NOT NULL,  
  `LastName` varchar(255) DEFAULT NULL,  
  `FirstName` varchar(255) DEFAULT NULL,  
  `BirthDate` varchar(50) DEFAULT NULL,  
  `Photo` varchar(255) DEFAULT NULL,  
  `Notes` varchar(255) DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

--

-- Dumping data for table `employees`

--

```
INSERT INTO `employees` (`EmployeeID`, `LastName`, `FirstName`, `BirthDate`, `Photo`, `Notes`)  
VALUES  
(1, 'Davolio', 'Nancy', '1968-12-08', 'EmpID1.pic', 'Education includes a BA in psychology.'),  
(2, 'Fuller', 'Andrew', '1952-02-19', 'EmpID2.pic', 'Andrew received his BTS commercial.'),  
(3, 'Leverling', 'Janet', '1963-08-30', 'EmpID3.pic', 'Janet has a BS degree in chemistry.'),  
(4, 'Peacock', 'Margaret', '1958-09-19', 'EmpID4.pic', 'Margaret holds a BA in English literature.'),  
(5, 'Buchanan', 'Steven', '1955-03-04', 'EmpID5.pic', 'Steven is a graduate of St. Andrews  
University.'),  
(6, 'Suyama', 'Michael', '1963-07-02', 'EmpID6.pic', 'Michael has studied Japanese literature.'),
```

```
(7, 'King', 'Robert', '1960-05-29', 'EmpID7.pic', 'Robert has an MA degree in history.'),
(8, 'Callahan', 'Laura', '1958-01-09', 'EmpID8.pic', 'Laura received a BA in psychology.'),
(9, 'Dodsworth', 'Anne', '1966-07-02', 'EmpID9.pic', 'Anne has a BA degree in Russian language.'),
(10, 'West', 'Adam', '1928-09-19', 'EmpID10.pic', 'An old chum');
```

```
-- -----
```

```
--
```

```
-- Table structure for table `orderdetails`
```

```
--
```

```
CREATE TABLE `orderdetails` (
```

```
  `OrderDetailID` int(11) NOT NULL,
```

```
  `OrderID` int(11) DEFAULT NULL,
```

```
  `ProductID` int(11) DEFAULT NULL,
```

```
  `Quantity` int(11) DEFAULT NULL
```

```
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```
--
```

```
-- Dumping data for table `orderdetails`
```

```
--
```

```
INSERT INTO `orderdetails` (`OrderDetailID`, `OrderID`, `ProductID`, `Quantity`) VALUES
```

```
(1, 10248, 29, 4),
```

```
(2, 10248, 38, 42),
```

```
(3, 10249, 71, 46),
```

```
(4, 10249, 67, 11),
```

```
(5, 10249, 49, 11),
```

```
(6, 10250, 32, 35),
```

```
(7, 10250, 30, 19),
```

```
(8, 10250, 23, 54),
```

```
(9, 10251, 74, 30),
```

```
(10, 10251, 56, 28),
```

```
(11, 10251, 59, 57),
```

```
(12, 10252, 36, 28),
```

```
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```

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(20, 10254, 32, 26),
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(49, 10264, 33, 48),

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(53, 10265, 63, 41),
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(56, 10266, 70, 49),
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(515, 10419, 66, 11),
(516, 10420, 40, 14),
(517, 10420, 20, 39),

```
(518, 10420, 31, 50);
```

```
-- -----
```

```
--
```

```
-- Table structure for table `orders`
```

```
--
```

```
CREATE TABLE `orders` (
```

```
  `OrderID` int(11) NOT NULL,
```

```
  `CustomerID` int(11) DEFAULT NULL,
```

```
  `EmployeeID` int(11) DEFAULT NULL,
```

```
  `OrderDate` varchar(50) DEFAULT NULL,
```

```
  `ShipperID` int(11) DEFAULT NULL
```

```
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```
--
```

```
-- Dumping data for table `orders`
```

```
--
```

```
INSERT INTO `orders` (`OrderID`, `CustomerID`, `EmployeeID`, `OrderDate`, `ShipperID`) VALUES
```

```
(10248, 90, 5, '7/4/1996', 3),
```

```
(10249, 81, 6, '7/5/1996', 1),
```

```
(10250, 34, 4, '7/8/1996', 2),
```

```
(10251, 84, 3, '7/8/1996', 1),
```

```
(10252, 76, 4, '7/9/1996', 2),
```

```
(10253, 34, 3, '7/10/1996', 2),
```

```
(10254, 14, 5, '7/11/1996', 2),
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```
-- -----
```

```
--
```

```
-- Table structure for table `products`
```

```
--
```

```
CREATE TABLE `products` (  
  `ProductID` int(11) NOT NULL,  
  `ProductName` varchar(100) NOT NULL,  
  `SupplierID` int(11) DEFAULT NULL,  
  `CategoryID` int(11) DEFAULT NULL,  
  `Unit` varchar(50) DEFAULT NULL,  
  `Price` decimal(10,2) DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```
--
```

```
-- Dumping data for table `products`
```

```
--
```

```
INSERT INTO `products` (`ProductID`, `ProductName`, `SupplierID`, `CategoryID`, `Unit`, `Price`)  
VALUES  
(1, 'Chais', 1, 1, '10 boxes x 20 bags', 18.00),  
(2, 'Chang', 1, 1, '24 - 12 oz bottles', 19.00),  
(3, 'Aniseed Syrup', 1, 2, '12 - 550 ml bottles', 10.00),  
(4, 'Chef Anton\'s Cajun Seasoning', 2, 2, '48 - 6 oz jars', 22.00),  
(5, 'Chef Anton\'s Gumbo Mix', 2, 2, '36 boxes', 21.35),  
(6, 'Grandma\'s Boysenberry Spread', 3, 2, '12 - 8 oz jars', 25.00),  
(7, 'Uncle Bob\'s Organic Dried Pears', 3, 7, '12 - 1 lb pkgs.', 30.00),  
(8, 'Northwoods Cranberry Sauce', 3, 2, '12 - 12 oz jars', 40.00),  
(9, 'Mishi Kobe Niku', 4, 6, '18 - 500 g pkgs.', 97.00),  
(10, 'Ikura', 4, 8, '12 - 200 ml jars', 31.00),  
(11, 'Queso Cabrales', 5, 4, '1 kg pkg.', 21.00),  
(12, 'Queso Manchego La Pastora', 5, 4, '10 - 500 g pkgs.', 38.00),  
(13, 'Konbu', 6, 8, '2 kg box', 6.00),
```

(14, 'Tofu', 6, 7, '40 - 100 g pkgs.', 23.25),
(15, 'Genen Shouyu', 6, 2, '24 - 250 ml bottles', 15.50),
(16, 'Pavlova', 7, 3, '32 - 500 g boxes', 17.45),
(17, 'Alice Mutton', 7, 6, '20 - 1 kg tins', 39.00),
(18, 'Carnarvon Tigers', 7, 8, '16 kg pkg.', 62.50),
(19, 'Teatime Chocolate Biscuits', 8, 3, '10 boxes x 12 pieces', 9.20),
(20, 'Sir Rodney\'s Marmalade', 8, 3, '30 gift boxes', 81.00),
(21, 'Sir Rodney\'s Scones', 8, 3, '24 pkgs. x 4 pieces', 10.00),
(22, 'Gustaf\'s Knäckebröd', 9, 5, '24 - 500 g pkgs.', 21.00),
(23, 'Tunnbröd', 9, 5, '12 - 250 g pkgs.', 9.00),
(24, 'Guaraná Fantástica', 10, 1, '12 - 355 ml cans', 4.50),
(25, 'NuNuCa Nuß-Nougat-Creme', 11, 3, '20 - 450 g glasses', 14.00),
(26, 'Gumbär Gummibärchen', 11, 3, '100 - 250 g bags', 31.23),
(27, 'Schoggi Schokolade', 11, 3, '100 - 100 g pieces', 43.90),
(28, 'Rössle Sauerkraut', 12, 7, '25 - 825 g cans', 45.60),
(29, 'Thüringer Rostbratwurst', 12, 6, '50 bags x 30 sausgs.', 123.79),
(30, 'Nord-Ost Matjeshering', 13, 8, '10 - 200 g glasses', 25.89),
(31, 'Gorgonzola Telino', 14, 4, '12 - 100 g pkgs', 12.50),
(32, 'Mascarpone Fabioli', 14, 4, '24 - 200 g pkgs.', 32.00),
(33, 'Geitost', 15, 4, '500 g', 2.50),
(34, 'Sasquatch Ale', 16, 1, '24 - 12 oz bottles', 14.00),
(35, 'Steeleye Stout', 16, 1, '24 - 12 oz bottles', 18.00),
(36, 'Inlagd Sill', 17, 8, '24 - 250 g jars', 19.00),
(37, 'Gravad lax', 17, 8, '12 - 500 g pkgs.', 26.00),
(38, 'Côte de Blaye', 18, 1, '12 - 75 cl bottles', 263.50),
(39, 'Chartreuse verte', 18, 1, '750 cc per bottle', 18.00),
(40, 'Boston Crab Meat', 19, 8, '24 - 4 oz tins', 18.40),
(41, 'Jack\'s New England Clam Chowder', 19, 8, '12 - 12 oz cans', 9.65),
(42, 'Singaporean Hokkien Fried Mee', 20, 5, '32 - 1 kg pkgs.', 14.00),
(43, 'Ipoh Coffee', 20, 1, '16 - 500 g tins', 46.00),
(44, 'Gula Malacca', 20, 2, '20 - 2 kg bags', 19.45),
(45, 'Røgede sild', 21, 8, '1k pkg.', 9.50),
(46, 'Spegesild', 21, 8, '4 - 450 g glasses', 12.00),
(47, 'Zaanse koeken', 22, 3, '10 - 4 oz boxes', 9.50),
(48, 'Chocolade', 22, 3, '10 pkgs.', 12.75),
(49, 'Maxilaku', 23, 3, '24 - 50 g pkgs.', 20.00),

(50, 'Valkoinen suklaa', 23, 3, '12 - 100 g bars', 16.25),
(51, 'Manjimup Dried Apples', 24, 7, '50 - 300 g pkgs.', 53.00),
(52, 'Filo Mix', 24, 5, '16 - 2 kg boxes', 7.00),
(53, 'Perth Pasties', 24, 6, '48 pieces', 32.80),
(54, 'Tourtière', 25, 6, '16 pies', 7.45),
(55, 'Pâté chinois', 25, 6, '24 boxes x 2 pies', 24.00),
(56, 'Gnocchi di nonna Alice', 26, 5, '24 - 250 g pkgs.', 38.00),
(57, 'Ravioli Angelo', 26, 5, '24 - 250 g pkgs.', 19.50),
(58, 'Escargots de Bourgogne', 27, 8, '24 pieces', 13.25),
(59, 'Raclette Courdavault', 28, 4, '5 kg pkg.', 55.00),
(60, 'Camembert Pierrot', 28, 4, '15 - 300 g rounds', 34.00),
(61, 'Sirop d'érable', 29, 2, '24 - 500 ml bottles', 28.50),
(62, 'Tarte au sucre', 29, 3, '48 pies', 49.30),
(63, 'Veggie-spread', 7, 2, '15 - 625 g jars', 43.90),
(64, 'Wimmers gute Semmelknödel', 12, 5, '20 bags x 4 pieces', 33.25),
(65, 'Louisiana Fiery Hot Pepper Sauce', 2, 2, '32 - 8 oz bottles', 21.05),
(66, 'Louisiana Hot Spiced Okra', 2, 2, '24 - 8 oz jars', 17.00),
(67, 'Laughing Lumberjack Lager', 16, 1, '24 - 12 oz bottles', 14.00),
(68, 'Scottish Longbreads', 8, 3, '10 boxes x 8 pieces', 12.50),
(69, 'Gudbrandsdalsost', 15, 4, '10 kg pkg.', 36.00),
(70, 'Outback Lager', 7, 1, '24 - 355 ml bottles', 15.00),
(71, 'Fløtemysost', 15, 4, '10 - 500 g pkgs.', 21.50),
(72, 'Mozzarella di Giovanni', 14, 4, '24 - 200 g pkgs.', 34.80),
(73, 'Röd Kaviar', 17, 8, '24 - 150 g jars', 15.00),
(74, 'Longlife Tofu', 4, 7, '5 kg pkg.', 10.00),
(75, 'Rhönbräu Klosterbier', 12, 1, '24 - 0.5 l bottles', 7.75),
(76, 'Lakkalikööri', 23, 1, '500 ml', 18.00),
(77, 'Original Frankfurter grüne Soße', 12, 2, '12 boxes', 13.00);

-- -----

--

-- Table structure for table `shippers`

--

CREATE TABLE `shippers` (

```
`ShipperID` int(11) NOT NULL,  
`ShipperName` varchar(100) NOT NULL,  
`Phone` varchar(50) DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

--

-- Dumping data for table `shippers`

--

```
INSERT INTO `shippers` (`ShipperID`, `ShipperName`, `Phone`) VALUES  
(1, 'Speedy Express', '(503) 555-9831'),  
(2, 'United Package', '(503) 555-3199'),  
(3, 'Federal Shipping', '(503) 555-9931');
```

-- -----

--

-- Table structure for table `suppliers`

--

```
CREATE TABLE `suppliers` (  
  `SupplierID` int(11) NOT NULL,  
  `SupplierName` varchar(100) NOT NULL,  
  `ContactName` varchar(100) DEFAULT NULL,  
  `Address` varchar(255) DEFAULT NULL,  
  `City` varchar(100) DEFAULT NULL,  
  `PostalCode` varchar(20) DEFAULT NULL,  
  `Country` varchar(100) DEFAULT NULL,  
  `Phone` varchar(20) DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

--

-- Dumping data for table `suppliers`

--

INSERT INTO `suppliers` (`SupplierID`, `SupplierName`, `ContactName`, `Address`, `City`,
`PostalCode`, `Country`, `Phone`) VALUES

(1, 'Exotic Liquid', 'Charlotte Cooper', '49 Gilbert St.', 'Londona', 'EC1 4SD', 'UK', '(171) 555-2222'),

(2, 'New Orleans Cajun Delights', 'Shelley Burke', 'P.O. Box 78934', 'New Orleans', '70117', 'USA', '(100) 555-4822'),

(3, 'Grandma Kelly\'s Homestead', 'Regina Murphy', '707 Oxford Rd.', 'Ann Arbor', '48104', 'USA', '(313) 555-5735'),

(4, 'Tokyo Traders', 'Yoshi Nagase', '9-8 Sekimai Musashino-shi', 'Tokyo', '100', 'Japan', '(03) 3555-5011'),

(5, 'Cooperativa de Quesos \'Las Cabras\'', 'Antonio del Valle Saavedra', 'Calle del Rosal 4', 'Oviedo', '33007', 'Spain', '(98) 598 76 54'),

(6, 'Mayumi\'s', 'Mayumi Ohno', '92 Setsuko Chuo-ku', 'Osaka', '545', 'Japan', '(06) 431-7877'),

(7, 'Pavlova, Ltd.', 'Ian Devling', '74 Rose St. Moonie Ponds', 'Melbourne', '3058', 'Australia', '(03) 444-2343'),

(8, 'Specialty Biscuits, Ltd.', 'Peter Wilson', '29 King\'s Way', 'Manchester', 'M14 GSD', 'UK', '(161) 555-4448'),

(9, 'PB Knäckebröd AB', 'Lars Peterson', 'Kaloadagatan 13', 'Göteborg', 'S-345 67', 'Sweden', '031-987 65 43'),

(10, 'Refrescos Americanas LTDA', 'Carlos Diaz', 'Av. das Americanas 12.890', 'São Paulo', '5442', 'Brazil', '(11) 555 4640'),

(11, 'Heli Süßwaren GmbH & Co. KG', 'Petra Winkler', 'Tiergartenstraße 5', 'Berlin', '10785', 'Germany', '(010) 9984510'),

(12, 'Plutzer Lebensmittelgroßmärkte AG', 'Martin Bein', 'Bogenallee 51', 'Frankfurt', '60439', 'Germany', '(069) 992755'),

(13, 'Nord-Ost-Fisch Handelsgesellschaft mbH', 'Sven Petersen', 'Frahmredder 112a', 'Cuxhaven', '27478', 'Germany', '(04721) 8713'),

(14, 'Formaggi Fortini s.r.l.', 'Elio Rossi', 'Viale Dante, 75', 'Ravenna', '48100', 'Italy', '(0544) 60323'),

(15, 'Norske Meierier', 'Beate Vileid', 'Hatlevegen 5', 'Sandvika', '1320', 'Norway', '(0)2-953010'),

(16, 'Bigfoot Breweries', 'Cheryl Saylor', '3400 - 8th Avenue Suite 210', 'Bend', '97101', 'USA', '(503) 555-9931'),

(17, 'Svensk Sjöföda AB', 'Michael Björn', 'Brovallavägen 231', 'Stockholm', 'S-123 45', 'Sweden', '08-123 45 67'),

(18, 'Aux joyeux ecclésiastiques', 'Guylène Nodier', '203, Rue des Francs-Bourgeois', 'Paris', '75004', 'France', '(1) 03.83.00.68'),

(19, 'New England Seafood Cannery', 'Robb Merchant', 'Order Processing Dept. 2100 Paul Revere Blvd.', 'Boston', '02134', 'USA', '(617) 555-3267'),
(20, 'Leka Trading', 'Chandra Leka', '471 Serangoon Loop, Suite #402', 'Singapore', '0512', 'Singapore', '555-8787'),
(21, 'Lyngbysild', 'Niels Petersen', 'Lyngbysild Fiskebakken 10', 'Lyngby', '2800', 'Denmark', '43844108'),
(22, 'Zaanse Snoepfabriek', 'Dirk Luchte', 'Verkoop Rijnweg 22', 'Zaandam', '9999 ZZ', 'Netherlands', '(12345) 1212'),
(23, 'Karkki Oy', 'Anne Heikkonen', 'Valtakatu 12', 'Lappeenranta', '53120', 'Finland', '(953) 10956'),
(24, 'G'day, Mate', 'Wendy Mackenzie', '170 Prince Edward Parade Hunter's Hill', 'Sydney', '2042', 'Australia', '(02) 555-5914'),
(25, 'Ma Maison', 'Jean-Guy Lauzon', '2960 Rue St. Laurent', 'Montréal', 'H1J 1C3', 'Canada', '(514) 555-9022'),
(26, 'Pasta Buttini s.r.l.', 'Giovanni Giudici', 'Via dei Gelsomini, 153', 'Salerno', '84100', 'Italy', '(089) 6547665'),
(27, 'Escargots Nouveaux', 'Marie Delamare', '22, rue H. Voiron', 'Montceau', '71300', 'France', '85.57.00.07'),
(28, 'Gai pâturage', 'Eliane Noz', 'Bat. B 3, rue des Alpes', 'Annecy', '74000', 'France', '38.76.98.06'),
(29, 'Forêts d'érables', 'Chantal Goulet', '148 rue Chasseur', 'Ste-Hyacinthe', 'J2S 7S', 'Canada', NULL);

--

-- Indexes for dumped tables

--

--

-- Indexes for table `categories`

--

ALTER TABLE `categories`

ADD PRIMARY KEY (`CategoryID`);

--

-- Indexes for table `customers`

--


```
ALTER TABLE `customers`  
  ADD PRIMARY KEY (`CustomerID`);  
  
--  
-- Indexes for table `employees`  
--  
ALTER TABLE `employees`  
  ADD PRIMARY KEY (`EmployeeID`);  
  
--  
-- Indexes for table `orderdetails`  
--  
ALTER TABLE `orderdetails`  
  ADD PRIMARY KEY (`OrderDetailID`),  
  ADD KEY `OrderID` (`OrderID`),  
  ADD KEY `ProductID` (`ProductID`);  
  
--  
-- Indexes for table `orders`  
--  
ALTER TABLE `orders`  
  ADD PRIMARY KEY (`OrderID`);  
  
--  
-- Indexes for table `products`  
--  
ALTER TABLE `products`  
  ADD PRIMARY KEY (`ProductID`),  
  ADD KEY `SupplierID` (`SupplierID`),  
  ADD KEY `CategoryID` (`CategoryID`);  
  
--  
-- Indexes for table `shippers`  
--  
ALTER TABLE `shippers`  
  ADD PRIMARY KEY (`ShipperID`);
```

```
--  
-- Indexes for table `suppliers`  
--  
ALTER TABLE `suppliers`  
  ADD PRIMARY KEY (`SupplierID`);  
  
--  
-- AUTO_INCREMENT for dumped tables  
--  
--  
--  
-- AUTO_INCREMENT for table `shippers`  
--  
ALTER TABLE `shippers`  
  MODIFY `ShipperID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT=4;  
  
--  
-- AUTO_INCREMENT for table `suppliers`  
--  
ALTER TABLE `suppliers`  
  MODIFY `SupplierID` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT=30;  
  
--  
-- Constraints for dumped tables  
--  
--  
--  
-- Constraints for table `orderdetails`  
--  
ALTER TABLE `orderdetails`  
  ADD CONSTRAINT `orderdetails_ibfk_1` FOREIGN KEY (`OrderID`) REFERENCES `orders`  
  (`OrderID`),  
  ADD CONSTRAINT `orderdetails_ibfk_2` FOREIGN KEY (`ProductID`) REFERENCES `products`  
  (`ProductID`);
```

```
--
-- Constraints for table `products`
--
ALTER TABLE `products`
  ADD CONSTRAINT `products_ibfk_1` FOREIGN KEY (`SupplierID`) REFERENCES `suppliers`
  (`SupplierID`),
  ADD CONSTRAINT `products_ibfk_2` FOREIGN KEY (`CategoryID`) REFERENCES `categories`
  (`CategoryID`);
COMMIT;

/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
```

1.ตัวนักศึกษาคือ ลูกค้า มาซื้อสินค้า 5 รายการ (ไมโล กล่องละ 10 บาท) ในระบบ product อีก 4 จะจัดเก็บข้อมูลอย่างไร

1. Customer 1 ระเบียบ

Insert into customer (customer, CustomerName) Values('92',ศิริประภา)

```
UPDATE `customers` SET `City` = 'เกาะคา', `Country` = 'ไทย' WHERE `customers`.`CustomerID` = 92;
```



ผลลัพธ์

✓ 1 row affected.

```
UPDATE `customers` SET `City` = 'ไมครา', `Country` = 'ไทย' WHERE `customers`.`CustomerID` = 92;
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 24 (92 total, Query took 0.0005 seconds.) [CustomerID: 92... - 68...]

```
SELECT * FROM `customers` ORDER BY `CustomerID` DESC
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> ☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: PRIMARY (DESC) v

Extra options

	CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
☐ Edit ☐ Copy ☐ Delete	92	ศิริประภา หอระดี	ศิริประภา หอระดี	3/1 ม.3 ต.วังพร้าว อ.เกาะคา จ.ลำปาง	เกาะคา	52130	ไทย

2. Product 1 ระบุเป็น

```
INSERT INTO `products` (`ProductID`, `ProductName`, `SupplierID`, `CategoryID`, `Unit`, `Price`) VALUES ('78', 'ไมโล', '14', '1', '500ml', '10');
```

```
SELECT * FROM `products` ORDER BY `ProductID` DESC
```

Preview SQL

```
INSERT INTO `products` (`ProductID`, `ProductName`, `SupplierID`,  
`CategoryID`, `Unit`, `Price`) VALUES ('78', 'ไมโล', '14', '1', '500ml', '10')
```

Close

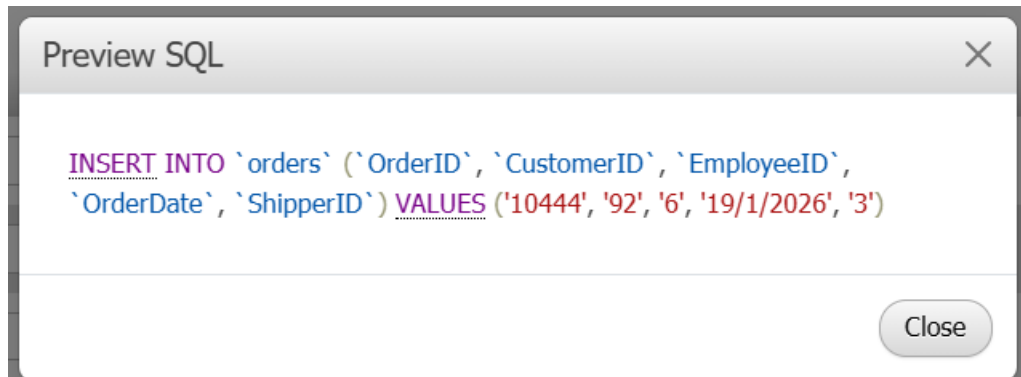
ผลลัพธ์

	ProductID	ProductName	SupplierID	CategoryID	Unit	Price
☐ Edit ☐ Copy ☐ Delete	78	ไมโล	14	1	500ml	10.00

3. Order 1 ระเบียน

```
INSERT INTO `orders` (`OrderID`, `CustomerID`, `EmployeeID`, `OrderDate`, `ShipperID`) VALUES ('10444', '92', '11', '19/1/2026', '3');
```

```
SELECT * FROM `orders` ORDER BY `OrderID` DESC
```



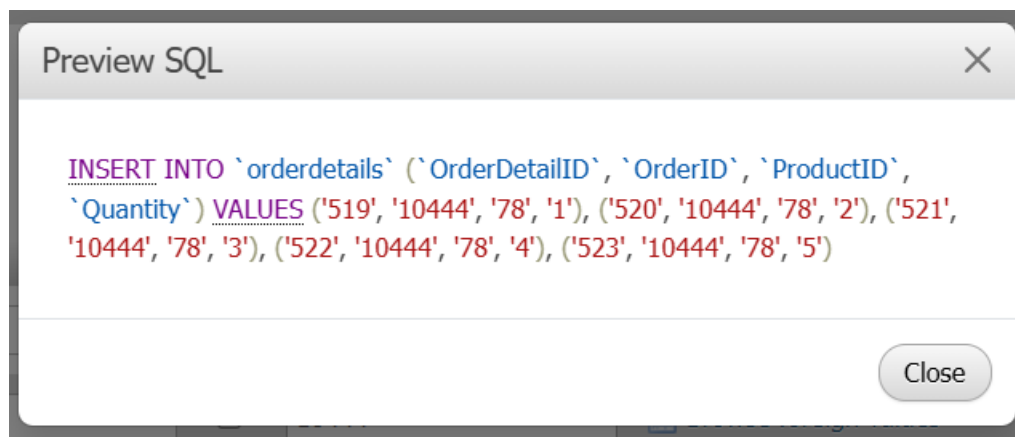
ผลลัพธ์

	OrderID	CustomerID	EmployeeID	OrderDate	ShipperID
☐ Edit Copy Delete	10444	92	11	19/1/2026	3

4. Orderdetail 5 ระเบียน

```
INSERT INTO `orderdetails` (`OrderDetailID`, `OrderID`, `ProductID`, `Quantity`) VALUES ('519', '10444', '78', '1'), ('520', '10444', '78', '1'), ('521', '10444', '78', '1'), ('522', '10444', '78', '1'), ('523', '10444', '78', '1');
```

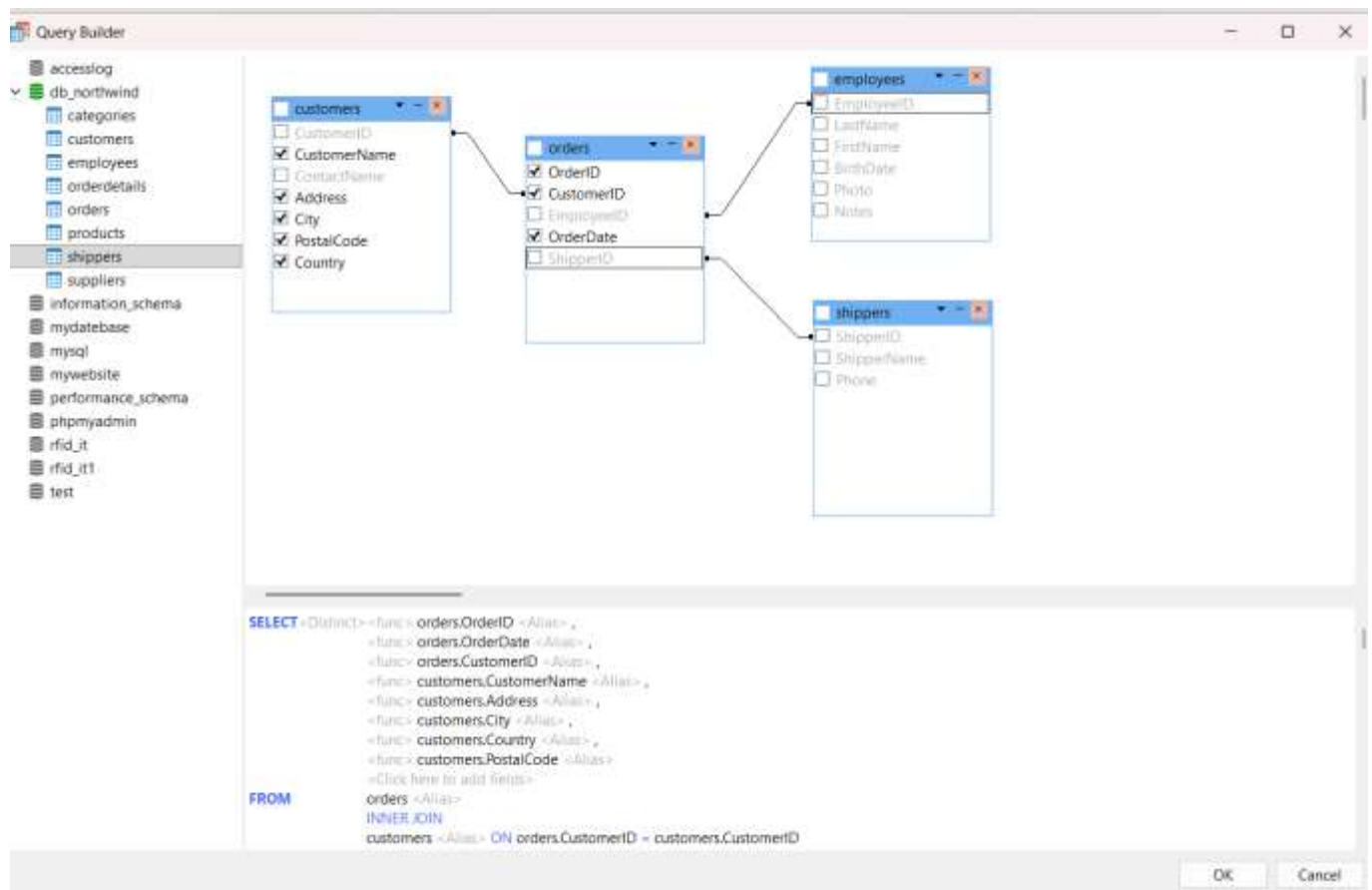
```
SELECT * FROM `orderdetails` ORDER BY `OrderDetailID` DESC
```



ผลลัพธ์

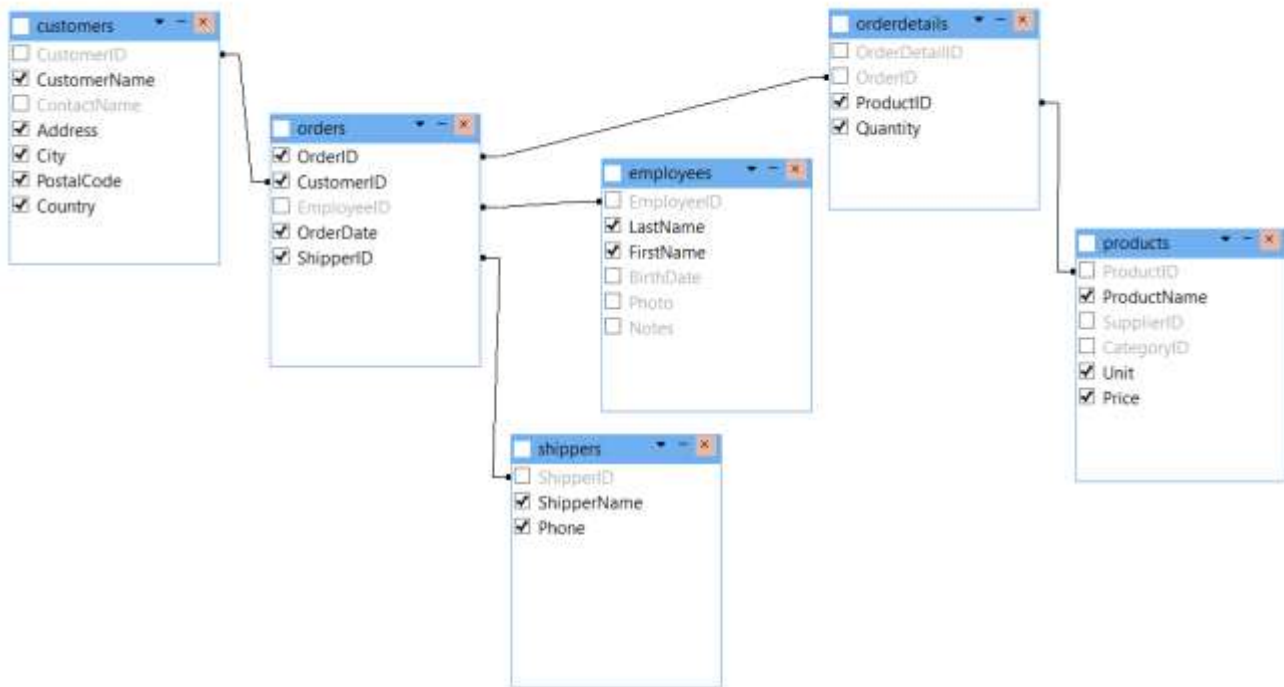
				OrderDetailID	1	OrderID	ProductID	Quantity
	Edit	Copy	Delete	523	10444	78	1	
	Edit	Copy	Delete	522	10444	78	1	
	Edit	Copy	Delete	521	10444	78	1	
	Edit	Copy	Delete	520	10444	78	1	
	Edit	Copy	Delete	519	10444	78	1	

ความสัมพันธ์ระหว่าง Order กับ OrderDetails



เพิ่ม total

```
SELECT
orders.OrderID,
orders.OrderDate,
orders.CustomerID,
customers.CustomerName,
customers.Address,
customers.City,
customers.Country,
customers.PostalCode,
employees.FirstName,
employees.LastName,
orders.ShipperID,
shippers.ShipperName,
shippers.Phone,
orderdetails.ProductID,
products.ProductName,
orderdetails.Quantity,
products.Unit,
products.Price ,Unit * Price as total
FROM
orders
INNER JOIN customers ON orders.CustomerID = customers.CustomerID
INNER JOIN employees ON employees.EmployeeID = orders.EmployeeID
INNER JOIN shippers ON orders.ShipperID = shippers.ShipperID
INNER JOIN orderdetails ON orderdetails.OrderID = orders.OrderID
INNER JOIN products ON orderdetails.ProductID = products.ProductID
```



Save Query Builder Beautify SQL Text Export Result

db_northwind db_northwind Run Selected Stop Explain

```

13  shippers.ShipperName,
14  shippers.Phone,
15  orderdetails.ProductID,
16  products.ProductName,
17  orderdetails.Quantity,
18  products.Unit,
19  products.Price ,Unit * Price as total
20  FROM
21  orders
22  INNER JOIN customers ON orders.CustomerID = customers.CustomerID
23  INNER JOIN employees ON employees.EmployeeID = orders.EmployeeID
24  INNER JOIN shippers ON orders.ShipperID = shippers.ShipperID
25  INNER JOIN orderdetails ON orderdetails.OrderID = orders.OrderID
26  INNER JOIN products ON orderdetails.ProductID = products.ProductID
27
  
```

แสดงผลลัพธ์

PostalCode	FirstName	LastName	ShipperID	ShipperName	Phone	ProductID	ProductName	Quantity	Unit	Price	total
05634-030	Michael	Suyama	1	Speedy Express	(503) 555-9831	71	Filetmysost	46	10 - 500 g pkgs.	21.5	215
05634-030	Michael	Suyama	1	Speedy Express	(503) 555-9831	67	Laughing Lumberjack	11	24 - 12 oz bottles	14	336
05634-030	Michael	Suyama	1	Speedy Express	(503) 555-9831	49	Maxilaku	11	24 - 50 g pkgs.	20	480
69004	Janet	Leverling	1	Speedy Express	(503) 555-9831	74	Longlife Tofu	30	5 kg pkg.	10	30
69004	Janet	Leverling	1	Speedy Express	(503) 555-9831	56	Gnocchi di nonna Alia	28	24 - 250 g pkgs.	38	912
69004	Janet	Leverling	1	Speedy Express	(503) 555-9831	59	Raclette Courdavault	57	5 kg pkg.	55	275
8010	Nancy	Davolio	1	Speedy Express	(503) 555-9831	27	Schoggi Schokolade	43	100 - 100 g pieces	43.9	4390
8010	Nancy	Davolio	1	Speedy Express	(503) 555-9831	6	Grandma's Boysenber	26	12 - 8 oz jars	25	300
8010	Nancy	Davolio	1	Speedy Express	(503) 555-9831	33	Geitost	23	500 g	2.5	1250
99508	Margaret	Peacock	1	Speedy Express	(503) 555-9831	41	Jack's New England Cl	43	12 - 12 oz cans	9.65	115.80000000000001
99508	Margaret	Peacock	1	Speedy Express	(503) 555-9831	59	Raclette Courdavault	49	5 kg pkg.	55	275
99508	Margaret	Peacock	1	Speedy Express	(503) 555-9831	47	Zaanse koeken	13	10 - 4 oz boxes	9.5	95
67000	Andrew	Fuller	1	Speedy Express	(503) 555-9831	77	Original Frankfurter gr	54	12 boxes	13	156

การสร้าง view มุมมองของผู้ใช้

```
Create VIEW vOrder_Bill as
SELECT
orders.OrderID,
orders.OrderDate,
customers.CustomerName,
customers.Address,
customers.City,
customers.PostalCode,
customers.Country,
orders.CustomerID,
orders.EmployeeID,
orders.ShipperID,
shippers.ShipperName,
shippers.Phone,
employees.FirstName,
employees.LastName,
orderdetails.ProductID,
orderdetails.Quantity,
products.ProductName,
products.Unit,
products.Price,
Unit * Price AS total
FROM
orders
INNER JOIN customers ON customers.CustomerID = orders.CustomerID
INNER JOIN employees ON orders.EmployeeID = employees.EmployeeID
INNER JOIN shippers ON orders.ShipperID = shippers.ShipperID
INNER JOIN orderdetails ON orderdetails.OrderID = orders.OrderID
INNER JOIN products ON orderdetails.ProductID = products.ProductID
WHERE orders.OrderID = '10444'
```

เรียกใช้

```
Select * form vOrder_Bill Where ordered = '10444'
```

```

1 Create VIEW vOrder_Bill as
2 SELECT
3 orders.OrderID,
4 orders.OrderDate,
5 customers.CustomerName,
6 customers.Address,
7 customers.City,
8 customers.PostalCode,
9 customers.Country,
10 orders.CustomerID,
11 orders.EmployeeID,
12 orders.ShipperID,
13 shippers.ShipperName,
14 shippers.Phone,
15 employees.FirstName,
16 employees.LastName,
17 orderdetails.ProductID,
18 orderdetails.Quantity,
19 products.ProductName,

```

Message Profile Status

```

orderdetails.ProductID,
orderdetails.Quantity,
products.ProductName,
products.Unit,
products.Price,
Unit * Price AS total
FROM
orders
INNER JOIN customers ON customers.CustomerID = orders.CustomerID
INNER JOIN employees ON orders.EmployeeID = employees.EmployeeID
INNER JOIN shippers ON orders.ShipperID = shippers.ShipperID
INNER JOIN orderdetails ON orderdetails.OrderID = orders.OrderID
INNER JOIN products ON orderdetails.ProductID = products.ProductID
WHERE orders.OrderID = '10444'
> OK
> Time: 0.004s

```

ผลลัพธ์

OrderID	OrderDate	CustomerName	Address	City	PostalCode	Country	CustomerID	EmployeeID	ShipperID	ShipperName	Phone	FirstName
10444	19/1/2026	บริษัท หน่อเตี	3/1 ม.3 คลองฟ้า อ.เกาะมหา		52130	ไทย	92	6	3	Federal Shipping	(503) 555-9931	Michael
10444	19/1/2026	บริษัท หน่อเตี	3/1 ม.3 คลองฟ้า อ.เกาะมหา		52130	ไทย	92	6	3	Federal Shipping	(503) 555-9931	Michael
10444	19/1/2026	บริษัท หน่อเตี	3/1 ม.3 คลองฟ้า อ.เกาะมหา		52130	ไทย	92	6	3	Federal Shipping	(503) 555-9931	Michael
10444	19/1/2026	บริษัท หน่อเตี	3/1 ม.3 คลองฟ้า อ.เกาะมหา		52130	ไทย	92	6	3	Federal Shipping	(503) 555-9931	Michael

1.นักศึกษา ซ้อนสินค้าเพิ่มอีก 3 รายการ และมีไมโล (Order กับ OrderDetail) จะได้OrderID ใหม่ 10445

```

INSERT INTO `orders` (`OrderID`, `CustomerID`, `EmployeeID`, `OrderDate`, `ShipperID`) VALUES ('10445', '92', '6', '19/1/2026', '3');

```

OrderID	CustomerID	EmployeeID	OrderDate	ShipperID
10445	92	6	19/1/2026	3

```
INSERT INTO `orderdetails` (`OrderDetailID`, `OrderID`, `ProductID`, `Quantity`) VALUES ('524', '10445', '78', '1'), ('525', '10445', '78', '1'), ('526', '10445', '78', '1');
```

OrderDetailID	OrderID ▼ 1	ProductID	Quantity
525	10445	78	1
526	10445	78	1
524	10445	78	1

2.แสดง vOrder_Bill ของOrderID = 10444 กับ 10445 เรียงจาก OrderID มากไปหาน้อย